Assignment 5 - Emerging Models Research Report

Model Selection: Qwen2.5-VL-7B and DeepSeek-Coder-6.7B

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Subject: Generative AI Model Development

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1. Introduction

Generative artificial intelligence has advanced rapidly in recent years, with models capable of producing text, images, code, and multimodal outputs. These models are increasingly being used in real-world applications, including creative tools and enterprise automation systems. Among the most recent advancements, two major directions in model development have emerged: multimodal understanding and domain-specific optimization.

This report analyzes two recent generative AI models that represent these directions: **Qwen2.5-VL-7B**, a vision-language model capable of interpreting images and text, and **DeepSeek-Coder-6.7B**, a specialized large language model designed for programming tasks. **Qwen2.5-VL-7B** advances multimodal perception by combining dynamic resolution processing and unified vision-language reasoning. **DeepSeek-Coder-6.7B**, on the other hand, is focused on software development and includes training strategies specifically designed for code generation and completion. This report compares these models to highlight the differences between general-purpose multimodal systems and specialized domain-focused models, as well as their technical innovations, industry impact, and practical applications.

2. Technical Overview

2.1 Qwen2.5-VL-7B Architecture

Qwen2.5-VL-7B is a multimodal vision-language model designed to handle both visual and textual data within a single architecture. The model combines a vision encoder and a transformer-based language model, allowing it to interpret images and produce natural language responses. This architecture allows the model to answer visual questions, describe images, analyze documents, and interpret charts.

The model introduces several technical innovations aimed at increasing the efficiency of multimodal reasoning. One significant innovation is dynamic resolution processing, which allows the model to process images of different sizes without forcing them to a fixed resolution. This improves visual fidelity while reducing unnecessary computational overhead. Furthermore, Qwen2.5-VL uses multimodal positional encoding techniques to represent spatial relationships in images alongside textual tokens within the transformer architecture.

The system also employs attention mechanisms that have been optimized to handle visual patches more efficiently. Instead of paying full attention to all visual tokens, localized attention windows are used to reduce computational complexity while maintaining important **spatial relationships**. These architectural improvements allow the model to maintain strong performance across tasks such as document understanding and chart analysis while remaining computationally efficient. Benchmark evaluations show strong multimodal reasoning abilities. The model performs particularly well on visual question answering and chart interpretation tasks, demonstrating its ability to extract structured information from visually represented data.

Key Specifications

Component	Description
Model Type	Vision-Language Transformer
Parameters	7 billion
Input Modalities	Images and text
Context Window	Up to 128K tokens
Key Innovation	Dynamic resolution processing and multimodal positional encoding
Primary Tasks	Visual question answering, chart analysis, document understanding

2.2 DeepSeek-Coder-6.7B Architecture

DeepSeek-Coder-6.7B is a specialized generative AI model created specifically for programming tasks. DeepSeek-Coder, unlike general-purpose language models, is focused on code understanding and generation, allowing it to help developers write, debug, and explain software programs.

The model is built with a transformer-based decoder architecture, which is common among modern large language models. However, its training data and objectives are designed specifically for software development tasks. A large portion of the training dataset is made up of source code extracted from publicly available repositories, as well as technical documentation and programming-related text.

One of the most significant innovations introduced in DeepSeek-Coder is the **Fill-in-the-Middle** (FIM) training strategy. Traditional language models generate text from left to right, but programming frequently requires code completion within existing structures. **FIM** training allows the model to generate missing sections of code between existing segments, making it ideal for real-world development tasks.

Another key feature is repository-level training, which analyzes code in the context of entire software projects rather than individual files. This enables the model to better understand the relationships between various components of a codebase, enhancing its ability to generate coherent and context-aware code suggestions.

Key Specifications

Component	Description
Model Type	Code-focused transformer language model
Parameters	6.7 billion
Architecture	Decoder-only transformer
Context Window	Up to 16K tokens
Training Data	Large-scale code datasets and technical documentation
Key Innovation	Fill-in-the-Middle code completion

2.3 Comparative Analysis

Although both models use transformer architecture, they are optimized for different tasks. **Qwen2.5-VL-7B** focuses on multimodal reasoning, which enables it to interpret both visual and textual data. **DeepSeek-Coder-6.7B**, on the other hand, focuses on software development tasks where specialized training leads to improved performance in code generation and debugging.

Dimension	**Qwen2.5-VL-7B**	**DeepSeek-Coder-6.7B**
Primary Domain	Vision-language multimodal tasks	Code generation and programming
Input Types	Images and text	Text and source code
Architecture Focus	Multimodal reasoning	Code understanding
Context Window	Up to 128K tokens	Up to 16K tokens
Optimization Goal	Visual perception and reasoning	Software development assistance

This comparison shows a key trend in modern generative AI development: while some models aim to become general-purpose multimodal systems, others are specialized models optimized for specific professional tasks.

3. Industry Impact

Generative AI models are rapidly transforming multiple industries by automating tasks that previously required human expertise. Vision-language models, such as **Qwen2.5-VL-7B**, provide powerful capabilities for interpreting visual data and combining it with language understanding. This technology has significant implications for industries such as finance, healthcare, manufacturing, and enterprise data processing.

Charts, reports, and dashboards, for example, are used to communicate large amounts of information in finance. These visual documents can be automatically analyzed by vision-language models, resulting in summaries and insights. This helps analysts to quickly interpret trends and identify anomalies in financial reports. Similarly, businesses that handle large amounts of documentation can use multimodal models to extract information from forms, invoices, and contracts.

Another important area of application is accessibility technology. Vision-language models can help visually impaired people by describing images, identifying objects in photographs, and summarizing graphical data like charts. These capabilities can improve digital accessibility and allow for greater participation in digital environments.

DeepSeek-Coder 6.7B has a significant impact on the software development ecosystem. AI-powered coding assistants are becoming increasingly common in modern development environments. These systems can generate code suggestions, identify bugs, and assist developers in understanding unfamiliar programming frameworks. Automating repetitive coding tasks allows developers to focus on higher-level problem solving and software design.

Furthermore, specialized coding models may help reduce entry barriers for people who want to learn programming. Educational platforms can use such models to provide interactive feedback, explain programming concepts, and guide students through coding exercises. This can lead to better learning outcomes and make programming education more accessible.

As generative AI models evolve, they are likely to play an increasingly important role in improving human productivity in a variety of professional fields.

4. Potential Applications

Both models support a variety of real-world applications depending on their domain specialization.

Model	Application	Implementation Example
Qwen2.5-VL-7B	Automated chart interpretation	Extract numerical data from charts and graphs
Qwen2.5-VL-7B	Document analysis	Extract key information from forms and reports
Qwen2.5-VL-7B	Accessibility tools	Generate image descriptions for visually impaired users
DeepSeek-Coder-6.7B	AI coding assistants	Generate and debug software code
DeepSeek-Coder-6.7B	Programming education	Provide automated feedback on student code

Combining multimodal and coding models is an emerging future direction. For example, a system could analyze a user interface screenshot using a vision-language model and automatically generate the corresponding code implementation using a coding model.

5. Practical Demonstration

To demonstrate the capabilities of **Qwen2.5-VL-7B**, a practical experiment was implemented in a Google Colab environment with GPU acceleration. The demonstration focused on two tasks that assessed the model's visual reasoning abilities.

Code available at: https://github.com/Venkata-Murari1711/Generative-AI-model-development

Case 1: Image-Based Question Answering

In the first experiment, the model examined a photograph of four people jumping on a beach at sunset. The model was asked to identify the number of people in the image, describe their poses, and interpret the overall mood. The model correctly identified the visual elements, such as the beach setting, silhouetted figures, and sunset lighting conditions. It also provided a meaningful interpretation of the scene by describing it as a happy or celebratory occasion.

```
system
You are a helpful assistant.
user
Extract the important visible information from this image.
Return:
- main title or heading
- important labels
- numbers or dates
- short summary
assistant
- **Main Title/Heading**: Joyful Sunset Celebration
- **Important Labels**: None
- **Numbers or Dates**: None
- **Short Summary**: A group of people is joyfully jumping on a beach at sunset, creating silhouettes against the colorful sky and ocean.
```

```
system
You are a helpful assistant.
user
Answer using only what you can see in the image.
1. What is this image about?
2. List 5 important visible details.
3. What is one likely purpose of this image?
4. Mention anything unclear or unreadable.
assistant
1. This image depicts a group of people jumping joyfully on a beach at sunset.

2. Important visible details:
   - Four silhouetted figures in mid-air, arms raised.
   - The beach setting with sand and water.
   - A sunset sky with warm hues transitioning to cooler tones.
   - The figures appear to be in a celebratory pose.
   - The horizon line where the ocean meets the sky.

3. One likely purpose of this image could be to convey a sense of freedom, happiness, or celebration.

4. There is no text or unclear elements in the image that need to be addressed.
```

Case 2: Chart and Graph Interpretation

The second experiment assessed the model's ability to interpret structured visual data. The model was given a chart that showed the number of units sold and total transactions over several months. The model correctly identified the chart type, summarized the data's trend, and estimated approximate values for the high and low points. This demonstrates the model's ability to extract information from visual data representations.

The demonstration showed that the model can process images at their native resolution and generate clear explanations within a few seconds using GPU acceleration.

```
system
You are a helpful assistant.
user
Read this chart and provide:
- a 3-bullet summary
- the most important trend
- one practical conclusion someone could make from this chart
assistant
### Summary:
1. **Units Sold**: The number of units sold shows a general upward trend, with significant increases in February and June.
2. **Total Transactions**: The total transaction value also shows an increasing trend, but it fluctuates more than the units sold.
3. **March Dip**: There is a noticeable dip in both units sold and total transactions in March.

### Most Important Trend:
The most important trend is the overall increase in units sold and total transactions over the period from January to June, with notable spi

### Practical Conclusion:
A practical conclusion someone could make from this chart is that there might be seasonal or promotional factors driving the increased sales
```

6. Limitations and Future Research

While both models have impressive capabilities, they also have significant limitations that must be considered when using them in real-world scenarios.

One limitation of vision-language models like Qwen2.5-VL-7B is that their performance is heavily reliant on image quality and visual clarity. If the input image has small text, low resolution, or ambiguous visual information, the model may make incorrect interpretations. In tasks like chart analysis, the model may estimate numerical values rather than extracting precise data points, which could lead to inaccuracies if the results are used to make decisions.

Similarly, coding models like DeepSeek-Coder-6.7B can produce technically correct code but still contain logical errors or security flaws. Automated code generation systems should therefore be used with caution, and human developers should thoroughly review generated code before deployment.

Another limitation concerns the transparency of training data. Large language models are frequently trained using massive datasets gathered from public sources, which may contain biases or copyrighted material. To address these issues, more research into dataset transparency, bias mitigation, and responsible AI development practices is needed.

Future research directions may concentrate on combining multimodal perception and domain-specific reasoning. For example, future models may combine visual understanding with coding capabilities, allowing systems to analyze graphical user interfaces and automatically generate corresponding software implementations. Such advancements may increase the role of generative AI in software engineering and data analysis.

7. Conclusion

This report looked at two emerging generative AI models that represent different directions in artificial intelligence research: Qwen2.5-VL-7B and DeepSeek-Coder-6.7B. Both models are based on transformer architectures, but they are optimized for different tasks. Qwen2.5-VL-7B focuses on multimodal understanding by combining visual perception with natural language reasoning, allowing for applications like image-based question answering and chart interpretation. DeepSeek-Coder-6.7B, on the other hand, is designed for programming tasks, with strong capabilities in code generation, debugging, and software assistance.

The practical demonstration in Google Colab confirmed Qwen2.5-VL-7B's ability to interpret both natural and structured visual data. The model was able to correctly identify visual elements, describe scenes, and summarize trends in charts. These results highlight the growing potential of multimodal AI systems in applications such as document processing, accessibility tools, and data analysis.

At the same time, the analysis of DeepSeek-Coder-6.7B demonstrates how domain-specific optimization can result in powerful software development tools. By focusing on programming data and specialized training techniques like fill-in-the-middle learning, the model can help developers with complex coding tasks.

Overall, the comparison between these two models shows an important trend in modern AI research: the coexistence of **general-purpose multimodal models and specialized domain-focused systems**. Future AI systems may combine these approaches to create more powerful and versatile tools that improve productivity and assist humans in making decisions across a wide range of fields.

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